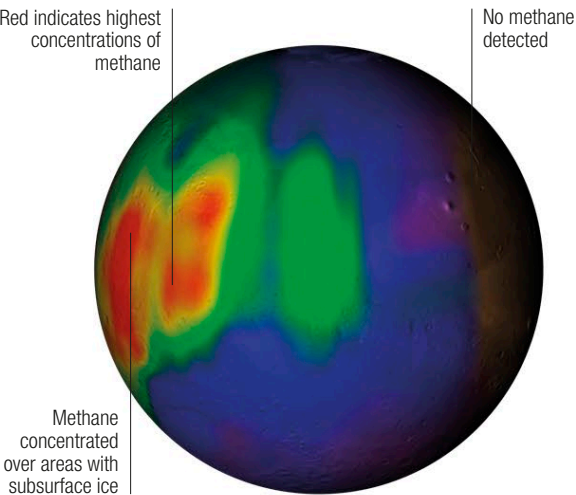
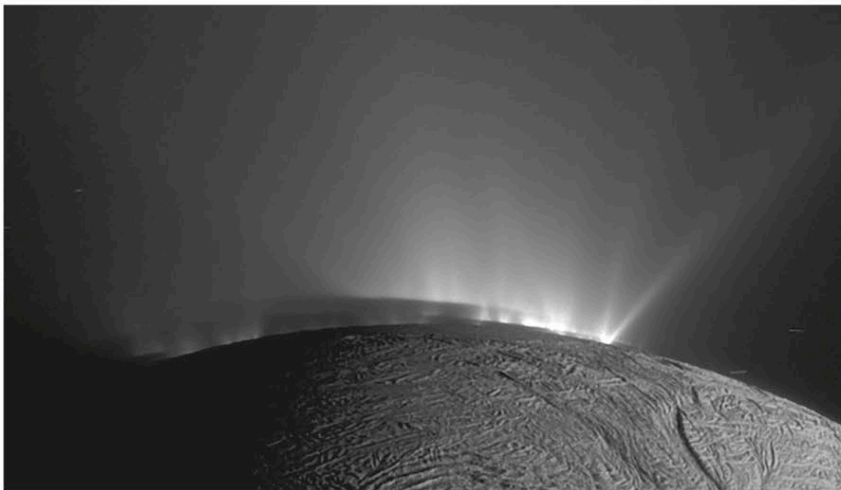


Signature of life

Any form of life must sustain itself through a series of chemical reactions known as the organism’s metabolism and, over time, this inevitably transforms the environment of the planet around it. For example, oxygen is a naturally reactive chemical that tends to get locked away as mineral compounds within rocks, so Earth would have no oxygen in its atmosphere if not for the evolution of life and the metabolic reactions of photosynthetic plants and algae over billions of years. Atmospheric oxygen is therefore a potential chemical biosignature for life on other worlds. Astronomers have already measured the atmospheres of a few exoplanets, but future telescopes should make this far more common.



◀ **Methane on Mars**
Methane is a gas that can only be produced by living microorganisms or active volcanism. It breaks down rapidly on exposure to sunlight, so the recent discovery of methane patches in the atmosphere of Mars raises intriguing questions about the Red Planet.



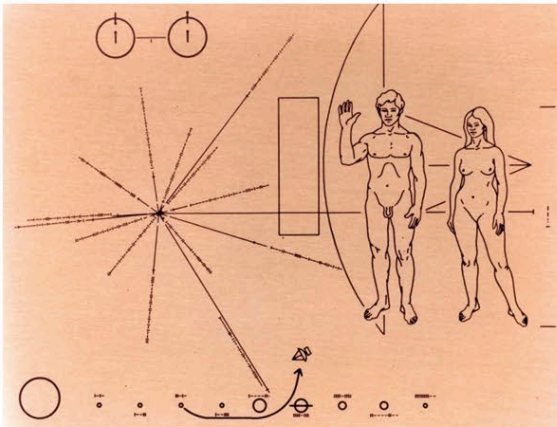
△ **Seas of Enceladus**
In 2005, NASA’s Cassini space probe discovered huge plumes of water ice erupting from Saturn’s small moon Enceladus, indicating a hidden ocean kept liquid by heating from powerful tidal forces. These make Enceladus one of the Solar System’s most likely habitats for life.

Estimates of the number of communicating civilizations in our galaxy range from many millions, to just one

Intelligent life

The Search for Extraterrestrial Intelligence (SETI) uses a variety of methods in the hope of tracking down evidence of intelligent aliens in the Universe. The most common is to scour the sky in search of artificial radio signals, but such signals are only likely to be found if aliens are deliberately beaming them toward us. Alternative approaches include looking for technosignatures (signs of technology), such as pollution in planetary atmospheres or even changes in the light output of stars created by alien engineering.

▷ **Messages in space**
Two space probes launched in the 1970s, Pioneer 10 (1972) and Pioneer 11 (1973), carried plaques with a pictogram message. This message was meant for any intelligent life that might intercept or recover one of the probes at some point in the future.



▷ **Drake's equation**
In 1961, SETI pioneer Frank Drake devised the Drake Equation, a way of assessing the number of civilizations that might be communicating by radio signals in the Milky Way at any one time.

KEY	
●	Drake equation 1961 estimates
●	Recent estimates

